

Multimedia Databases Exercises

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Exercise 2

Subject: Multimedia and Multimedia Databases - Initial Concepts II

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Task 1: Semantic Gap

- a) Refer to Figure 1. Based on this picture give examples for “low level features” and “high level features”. Following from this, how would you interpret the term “bridging the semantic gap”?
- b) Identify analogous “low” and “high level features” for an audio signal from a recording of a phone call.



Figure 1: Semantic Gap

Task 2: Components of MMDB

- a) Where does the difference between multimedia databases and “classical” databases lie? Consider in particular how the different components of databases (e.g. indexing, data model, query languages...) handle the data structures from Exercise 1.
- b) Consider a video-hosting platform similar to Youtube. Which components listed in a) are relevant to the functionalities ‘account’, ‘video playlist’, ‘video recommender’ and ‘comment section’.